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Project Proposal

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**Project Title: Smart Elderly Care AI - A Multimodal AI System for Proactive
Monitoring, Forecasting, and Care of Elderly Aged 60 and Above**

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PROJECT INFORMATION

Project acronym	Smart Elderly Care AI		
Project Title	Smart Elderly Care AI - A Multimodal AI System for Proactive Monitoring, Forecasting, and Care of Elderly Aged 60 and Above		
Start Date	21/08/2026	End Date	06/12/2026
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REVISION HISTORY

Version	Date	Comments	Author
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Table of Contents

1. Project Title	6
2. Team Members.....	6
3. Supervisor(s).....	6
4. Problem Statement.....	7
5. Survey / Existing Solution	8
6. Objectives and Scope	10
6.1 Objectives	10
6.1.1 Building a Multimodal Health Monitoring Ecosystem (ElderCare-AI)	10
6.1.2 In-scope	11
6.2 Scope not included	12
7. Key Features & Requirements.....	13
7.1 Key features	13
7.2 Requirement.....	13
7.2.1 Functional requirement.....	13
7.2.2 Non-Functional requirement	16
8. Constraints and Assumptions	18
8.1 Constraints	18
8.2 Assumption.....	19
9. Target Users / Stakeholders	20
9.1 Elderly Individuals (Primary Beneficiaries).....	20
9.2 Family Caregivers	21
9.3 System Administrators.....	21
10. Technology Stack	21
11.1 Scrum Process	22
11.2 Development Plan	23
12. System Architecture Overview	28
Figure 1. System context diagram	28
13. Potential Risks and Mitigation Strategies.....	29
14. Expected Outcomes / Deliverables.....	30
15. References	31

1. Project Title

Smart Elderly Care AI - A Multimodal AI System for Proactive Monitoring, Forecasting, and Care of Elderly Aged 60 and Above.

2. Team Members

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4. Problem Statement

What challenges do users/customers face?

Elderly individuals aged 60 and older may experience serious safety incidents while living alone, including falls, prolonged immobility, abnormal temperature, and other potentially dangerous events. These incidents may not be detected immediately when no caregiver is present. Current monitoring solutions have limitations: smartwatches are difficult for the elderly to use and have short battery lives (1–2 days), leading users to forget to charge or wear them; cameras compromise privacy by storing raw video in the cloud, suffer from high alert latency due to internet dependency, and have high false-alarm rates (failing to distinguish between sleeping and an actual fall). Notably, there is currently no system that integrates multiple sensors for cross-verification of incidents, nor is there a mechanism to automatically transmit video evidence and rapidly connect to family members or emergency services (115) when an accident occurs.

What is the impact of this issue?

Delays in detecting incidents cause elderly people to miss the “golden time” for emergency medical treatment, resulting in a high risk of death or permanent disabilities (traumatic brain injuries, hemiplegia, bone necrosis). For families, caregivers are constantly anxious and worried when working far away. For healthcare facilities, hospitals and emergency medical teams (115) face difficulties due to receiving information late and lacking historical vital-sign data (heart rate, SpO2, fall progression) before the patient arrives at the hospital, reducing the effectiveness of initial diagnosis and treatment. In addition, standalone monitoring devices frequently generate false alarms, causing unnecessary overload for the healthcare system and families.

Why did the team choose to solve this problem?

The team aims to apply Edge AI technology combined with Multimodal Sensor Fusion, including visual cameras, thermal matrix sensors, audio analysis, image analysis, and wearable devices. This solution helps accurately detect incidents, eliminate false alarms, protect the privacy of images at home, and immediately send alerts along with vital-sign data to family members, authorized persons (family

doctors), and the hospital emergency system (115 hotline), thereby optimizing medical intervention time and improving the quality of safety protection for elderly people.

5. Survey / Existing Solution

Criteria	Smart Elderly Care AI	SafelyYou	Vayyar Care	Sensi.AI
Development Environment	WEB Mobile App (Camera + Audio + Wearable)	Mobile App + Cloud (AI Chamber)	Web + Cloud Dashboard (Radar Sensor)	Mobile App + Cloud (Audio pods)
Sensor Method	Multimodal (full) – combining Camera (image), Audio (voice/noise), and Wearable (vital indicators, movement) in the same system	Camera-based – Recording when a fall is detected	Radar millimeta - wave – no cameras, no wearable devices; Through-wall/furniture sensors	Audio pods – no cameras, no wearable devices; Environmental Sound Analysis
Fall detection	Yes (cross confirmation between Camera + Wearable,	Yes (high accuracy, video fault verification)	Yes (real-time, privacy protection)	Indirect (inferred from fall sounds/cries for help, no image verification)

	reduce false alarm)			
AI Health Prediction/Assessment	Advanced (AI predicts health risks, combining visual, audio, and vital signs data from wearables)	Yes (Analyze fall data to create care insights)	Yes (analysis of movement trends, care needs)	Yes (early detection of signs of urinary infections, pneumonia, dementia, voice agitation)
Privacy Security	Medium (with camera + vital sign data, needs separate security/encryption mechanism)	Medium (archive of incident images/videos)	High (no camera, no image recognition)	High (sound only, "privacy-first" design)
Communicating with caregivers/families	Advanced (real-time alert via Web + App, synchronize multi-source data for relatives/authors)	Yes (send incident/alert video to the care team)	Yes (real-time notifications on the dashboard for employees)	Yes (send audio recordings, transcripts, recommendations for action)
Event chat group	Optional (Automatically create a real-time chat tool to	None (Communication and coordination	None (Communication and coordination	None (Information is communicated one-way from the organizer via

	coordinate, communicate and coordinate events)	usually take place through personal channels such as Messenger, Zalo)	usually take place through personal channels such as Messenger, Zalo)	notification or email)
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6. Objectives and Scope

6.1 Objectives

6.1.1 Building a Multimodal Health Monitoring Ecosystem (ElderCare-AI)

- **S (Specific):** Design and fully develop an integrated health monitoring system combining Edge AI (running on an Orange Pi 5 single-board computer), a cloud backend, and a mobile application (React Native). The system integrates four sensor channels—standard camera with audio, an AMG8833 thermal sensor, and a BLE vital-sign tracking wristband—to automatically detect falls, strokes, high fever, and respiratory distress.
- **M (Measurable):**
 - o Edge AI (YOLO) pose estimation inference speed reaches at least 30–35 frames per second.
 - o Response time from incident occurrence to emergency alert activation is under 3.0 seconds.
 - o Latency for MQTT alert packet transmission from the Edge Hub to the server and mobile application is under 500 ms.
 - o The system will be evaluated using at least 50 controlled test scenarios covering falls, immobility, abnormal temperature, abnormal heart rate, low SpO2, and multimodal combinations.
- **A (Achievable):** With a team of software engineers proficient in Python, FastAPI, React Native, and IoT/AI processing, and a 3.5-month

implementation timeline, completing the hardware-software prototype is entirely feasible.

- **R (Relevant):** The product directly addresses safety concerns for elderly individuals living alone, alleviates psychological burdens on families, and optimizes hospital emergency response times.
- **T (Time-bound):** Complete the entire development, system integration, and field testing phases prior to the final evaluation deadline.

6.1.2 In-scope Function

- Hardware-accelerated or optimized Edge AI model for 17-keypoint skeletal pose estimation and immobility detection(Orange Pi 5 using the integrated Rockchip RK3588S NPU).
- AI-based audio event classification for distress sounds, calls for help, screams, and significant impact sounds.
- Contactless thermal screening to estimate forehead surface temperature and identify abnormal temperature patterns.
- Data collection (heart rate, SpO2, and impact acceleration) from a BLE wearable device.
- Multimodal sensor integration engine designed to minimize false alarms.
- High-priority emergency push alert system that overrides silent mode on the caregiver's device.
- Automatic extraction of Short incident video clips extracted from a configurable circular media buffer around the detected event capturing the incident scene upon a "Red Alert" trigger.
- Dashboard for monitoring time-series vital sign data and one-touch SOS (115) emergency calling.

- **Live monitoring and playback:** Supports live camera streaming (Live View) and a timeline slider (Web/Mobile) to review the rolling 1–2 day video history recorded at the Hub.

User

- **Elderly User:** Beneficiary of automated safety monitoring, receiving localized voice reminders from the Hub when forgetting to wear the wristband.
- **Caregiver / Family Member:** Receives red alert SOS alarm sirens, views incident video clips, monitors real-time vital signs, and places 115 emergency calls.
- **Family Physician / Administrator:** Sets vital sign alert thresholds, reviews medical history logs, and configures hardware devices.

6.2 Scope not included

- **Deep Medical Hardware Intervention:** The system does not replace specialized intensive care unit (ICU) therapeutic equipment, such as mechanical ventilators, defibrillators, or 12-lead electrocardiogram (ECG) machines.
- **Direct Integration into Emergency 115 Telecommunications Infrastructure:** Emergency calls and location data dispatched to 115 emergency services are triggered through automated emergency speed-dial protocols via the caregiver's mobile application, without directly interfacing with national emergency dispatch core systems.
- **Legally Binding Automated Medical Diagnostics:** Vital sign and body temperature alerts function strictly as early-warning remote monitoring tools and do not provide diagnostic medical conclusions in place of certified physician evaluation.

- **Large-scale Commercialization:** The project focuses on research, system integration, and testing of a functional Prototype/Beta version, excluding recurring subscription billing models, mass commercial production, or e-commerce payment gateways.

7. Key Features & Requirements

7.1 Key features

- User Authentication and Role-Based Access Control (RBAC)
- Real-time vital signs monitoring dashboard (Heart rate, SpO2, Body temperature)
- Edge-AI Fall & Immobility Detection (AI identifies and detects fall events and prolonged motionless lying)
- Contactless forehead thermal fever screening
- Acoustic distress & impact sound classification (Detects Help/Scream calls and mechanical impacts)
- Multimodal Sensor Fusion to eliminate false alarms
- Critical Push Notification & Silence-override alarm siren
- Edge-buffered 5-second incident video clip auto-upload
- Live streaming & 24/48h interactive video timeline playback
- Historical incident review & Cloud media storage
- One-touch SOS Emergency 115 speed dial with auto-copied location
- Smart voice reminder for elderly users when forgetting the smart wristband
- Medical history time-series logging & report export

7.2 Requirement

7.2.1 Functional requirement

Order ID	Key Feature	Description
FR01	User Authentication & Authorization	Supports registration and login via Email/Password or Google OAuth2; issues fine-grained role permissions between primary Caregivers and authorized users (e.g., family physician).
FR02	Real-time Vital Signs Dashboard	Mobile and Web application displaying real-time heart rate (BPM), blood oxygen level (SpO2), and body temperature updated via WebSocket protocol.
FR03	Edge-AI Fall & Posture Detection	YOLO-Pose model running directly on edge hardware to recognize 17 skeletal keypoints, calculate spinal tilt angle, and detect fall events or motionless floor lying.
FR04	Contactless Thermal Mapping	The system shall collect 8×8 thermal matrix data from the AMG8833 infrared thermal sensor, identify the relevant forehead region, estimate forehead surface temperature, and detect abnormal temperature patterns for early-warning purposes.
FR05	Acoustic Distress Analysis	The system shall classify predefined acoustic events such as impacts, screams, calls for help, and selected abnormal breathing-related sounds.
FR06	Multimodal Sensor Fusion	Central processing unit concurrently fuses 4 data streams (Vision + Thermal + Audio + BLE Wristband) using decision matrices to eliminate

		false positives before triggering alarms.
FR07	Critical Push Notification	Upon detecting critical events, the backend server pushes highest-priority notifications (FCM High Priority), waking mobile screens and sounding max-volume sirens on caregiver devices.
FR08	Edge-Buffered Incident Clip	Local Hub maintains a 5-second circular ring buffer in RAM; upon red alert, extracts a 5s video clip with snapshot image and auto-uploads to Cloud Object Storage.
FR09	Live Streaming & 24/48h Timeline Playback	Caregivers can view live camera streams (via WebRTC/HLS) and drag the interactive playback timeline on Mobile/Web to review all daily activity recorded in the past 1–2 days (rolling buffer).
FR10	Historical Incident Review	Allows querying past incident logs, retrieving media URLs from database, and replaying video snapshots and evidence images stored on Cloud Storage.
FR11	One-Touch SOS Emergency 115	App integrates a single-tap emergency SOS button, enabling caregivers to place an immediate speed dial to Emergency Services (115) with auto-copied victim residential address.
FR12	Smart Voice Reminder	Home Hub plays localized friendly audio voice prompts in Vietnamese upon detecting elderly

		individuals leaving bed or moving without wearing their smart BLE wristband.
FR13	Medical History & Data Export	Stores time-series vital sign records in PostgreSQL database, allowing health history lookup and PDF/Excel health report export by week or month.

7.2.2 Non-Functional requirement

Order ID	Name	Description
NFR01	Performance & Latency	- The edge skeleton pose estimation speed must achieve at least 30 FPS. - The on-device fall detection latency must be less than 3.0 seconds. - The end-to-end alert message transmission latency via MQTT/FCM to caregiver mobile devices must be under 500 ms. - The live camera stream connection latency must be under 1.0 second; incident video clip playback load time must be under 2.0 seconds.
NFR02	Security & Privacy	- All AI vision inference processing must be executed locally on the Edge Hub, ensuring raw video streams are never continuously transmitted to the Internet.

		- Media uploads are strictly limited to authenticated incident images/short clips. - Media transmissions and WebRTC/HLS/MQTT connections must be encrypted via TLS/SSL and authenticated using JWT.
NFR03	Usability	- The mobile app (React Native) and Web Dashboard user interface (UI) need to be intuitive, clean, and newbie-friendly. - The search and filtering features in the system should be quick and easy to use, featuring interactive timeline sliders for historical video review.
NFR04	Reliability & False Alarm Filtering	- The home Edge Hub must operate continuously 24/7, with automatic service restoration (Systemd Daemon) upon power recovery without requiring manual operation by elderly users. - The system must maintain a True Positive Rate (TPR) $\geq 95\%$ and keep the False Positive Rate (FPR) $< 2\%$ using a multimodal Sensor Fusion cross-verification mechanism.

NFR05	Scalability & Storage Tiering	The backend system (FastAPI + EMQX Broker + Redis) is containerized via Docker and needs to be scalable to handle thousands of home Hubs without performance issues. - Tiered storage strategy: Local Hub storage (SD Card/USB) automatically overwrites in a circular (FIFO) loop after 24–48 hours; critical incident videos on Cloud Object Storage are automatically purged after 30–90 days.
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8. Constraints and Assumptions

8.1 Constraints

- **Time:** Time to complete the project within 3.5 months (from August 21, 2026 to December 06, 2026).
- **Budget:** The project cost does not exceed \$500 USD.

Category	Item	Cost
Hardware	Laptops/PCs for team (5 persons)	\$0
	Orange Pi 5 – Rockchip RK3588S	\$138
	Far-Infrared Thermal Sensor Matrix(AMG8833)	\$35
	IP Camera / USB Wide-Angle Camera Module	\$25
	BLE Smartband (Heart Rate, SpO2, Accelerometer)	\$30
	MicroSD High Endurance (64GB) + Official 5V/3A USB-C Power Supply	\$15

Software & AI	Open-source Frameworks (FastAPI, React Native, Next.js, Docker)	\$0 (open source)
	Edge AI Pretrained Models (YOLOv8n-Pose via, TFLite/ONNX Runtime YAMNet)	\$0 (open source)
	Database Engine (PostgreSQL, TimescaleDB, Redis)	\$0 (open source)
	Domain Name (.tech / .site for Web Dashboard & API)	\$1 / month
Hosting & Cloud	Cloud VPS / Server (Ubuntu Host for Backend & EMQX Broker)	\$12 / month
	Cloud Object Storage (MinIO / Firebase Storage for 5s Clips & Logs)	\$5 / month
Development Tools	IDEs & Editors (VS Code, Android Studio, DBeaver, MQTTX)	\$0
	Version Control (GitHub Free Tier / GitHub Team)	\$0
Management Tool	Jira Software (Atlassian Agile Free Plan for ≤10 users)	\$0
Communication	Google Meet, Discord, Zalo	\$0
Total	Hardware One-Time Cost + Operating Hosting Cost	~\$261 USD

- **Human resource:** 5 person

8.2 Assumption

- **Stable System Requirements:** The functional and safety monitoring requirements are clearly defined from the beginning with no major architectural pivots during development.

- **Sufficient Hardware Budget:** The allocated personal budget is adequate to procure the physical hardware prototype (Orange Pi 5 – Rockchip RK3588S, AMG8833, Camera, Smartband) by October 2026.
- **PC-First Simulation Validity:** The core algorithms developed and tested on PC (via webcam and mock sensor telemetry) will The team assumes that the core algorithms developed and tested on PC can be adapted to Orange Pi 5 through model optimization, ONNX conversion, RKNN conversion, and hardware-specific runtime configuration.
- **User Usability:** Caregivers and family members can easily navigate the mobile app/web dashboard to monitor health data and respond to SOS alarms without specialized technical training.
- **Reliable Operating Environment:** The home test environment provides a stable 2.4GHz/5GHz Wi-Fi connection, continuous AC power supply, and standard residential lighting for optimal camera inference.

9. Target Users / Stakeholders

9.1 Elderly Individuals (Primary Beneficiaries)

- **Target Group:** Seniors aged 60 and older living independently, individuals with cardiovascular issues, reduced mobility, or high risks of stroke and mechanical falls.
- **Key Benefits:** Passive 24/7 Safety Safeguard: Ambient and continuous protection without the burden of operating complex electronic devices.
- **Autonomous Emergency Detection:** Instantaneous detection of falls, immobility, and acute fevers without relying on manual panic buttons.
- **Strict In-Home Privacy:** Visual AI inference runs locally on the edge hub, eliminating private video streaming over the cloud. **Smart Voice Guidance:** Audio prompts from the hub automatically remind seniors to wear their biometric band.

9.2 Family Caregivers

- **Target Group:** Adult children, remote family members, or dedicated caregivers managing elderly relatives.
- **Key Benefits:** Remote Health Visibility: Real-time monitoring of vital signs (Heart Rate, SpO2, Body Temperature) via a dedicated mobile application.
- **Critical Incident Alerts:** High-priority siren notifications overriding silent mode, accompanied by a 5-second incident verification clip.
- **Rapid Emergency Coordination:** Integrated one-touch 115 emergency dialer with automatic location sharing.
- **Cost Efficiency:** Low-cost hardware deployment with zero recurring subscription maintenance fees.

9.3 System Administrators

- **Target Group:** Technical personnel responsible for operating, maintaining, and scaling the backend infrastructure.
- **Key Benefits:** Centralized Fleet Management: Real-time visibility into the connection status of distributed home hubs via Redis and MQTT.
- **Security and Data Integrity:** Centralized access control via JWT authentication and encrypted data transmission using TLS protocols.

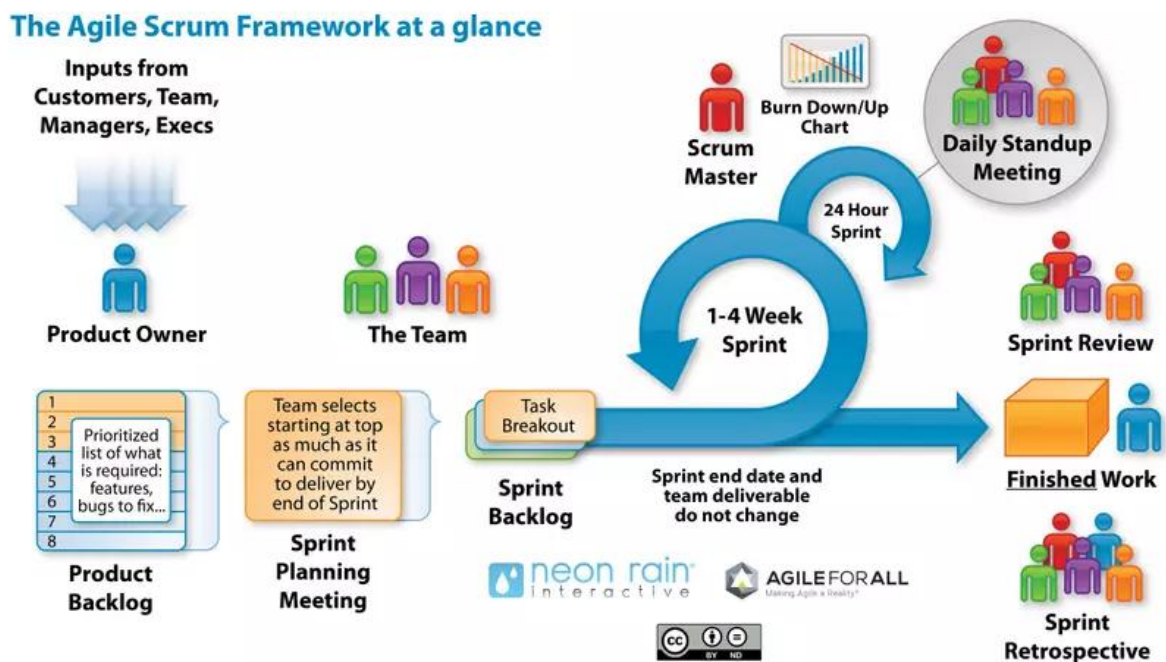
10. Technology Stack

- **Technology**
 - o Frontend: React.js / Next.js (Web), React Native (Mobile), TypeScript
 - o Backend: Python, FastAPI
 - o Cloud & Message Broker: Firebase (FCM), EMQX (MQTT Broker)
 - o Database: PostgreSQL, Redis
 - o AI/ML & Embedded: Python, YOLOv8-Pose, YAMNet
- **Development Tools**
 - o Tools: Visual Studio Code, Android Studio, Postman, Docker
 - o Database: PostgreSQL, Redis
- **Management Tools**

- o Tools: Git, GitHub, Jira

11. Methodology & Development Plan

11.1 Scrum Process



- Scrum is an iterative and incremental agile software development framework for managing software projects and product or application development.
- Scrum focuses on project management institutions where it is difficult to plan ahead.
- Mechanisms of empirical process control, where feedback loops that constitute the core management technique is used as opposed to traditional command-and-control management.
- Its approach to planning and managing projects is by bringing decision-making authority to the level of operation properties and certainties.

Benefit of the methodology:

- The project can respond easily to change.
- Problems are identified early.
- Customers get the most beneficial work first.
- Work done will better meet the customer's needs.
- Improved productivity.
- Ability to maintain a predictable schedule for delivery.

11.2 Development Plan

NO	Task Name	Duration	Start	Finish
		108 days	21/08/2026	06/12/2026
1.	Initial	7 days	21/08/2026	27/08/2026
1.1	Gathering Requirements (Elderly Fall & Health Monitoring)	4 days	21/08/2026	24/08/2026
1.2	Create Proposal Document & Hardware Feasibility	3 days	25/08/2026	27/08/2026
2.	Start Up	7 days	28/08/2026	03/09/2026
2.1	Project Kick-off Meeting & Setup Dev Environment	3 days	28/08/2026	30/08/2026
2.2	Create System Architecture & DB Design Document	4 days	31/08/2026	03/09/2026
3.	Development	93 days	03/09/2026	04/12/2026
3.1	Sprint 1 (Code Core: AI Fall Detection on PC & Backend Foundation)	25 days	03/09/2026	27/09/2026

3.1.1	Design Database Schema (PostgreSQL, TimescaleDB, Redis)	3 days	03/09/2026	05/09/2026
3.1.2	Design UX/UI (Vitals Dashboard, Live Stream, SOS Alert, Timeline Playback)	4 days	06/09/2026	09/09/2026
3.1.3	Develop YOLOv8-Pose Fall Detection on PC (Test via Webcam/Public Dataset)	6 days	10/09/2026	15/09/2026
3.1.4	Develop Backend APIs (Auth, User Profile, MQTT Gateway, WebSocket Hub)	5 days	16/09/2026	20/09/2026
3.1.5	Build Mock Sensor Generator (Simulate Heart Rate, SpO2 & Thermal Matrix)	3 days	21/09/2026	23/09/2026
3.1.6	Test AI Inference & Backend APIs on PC	2 days	24/09/2026	25/09/2026
3.1.7	Fix Bugs & Optimize Algorithm	2 days	26/09/2026	27/09/2026
3.2	Sprint 2 (Hardware Procurement, NPU Porting & Sensor Integration)	26 days	28/09/2026	23/10/2026
3.2.1	Purchase & Setup Physical Devices (Orange Pi 5,	4 days	28/09/2026	01/10/2026

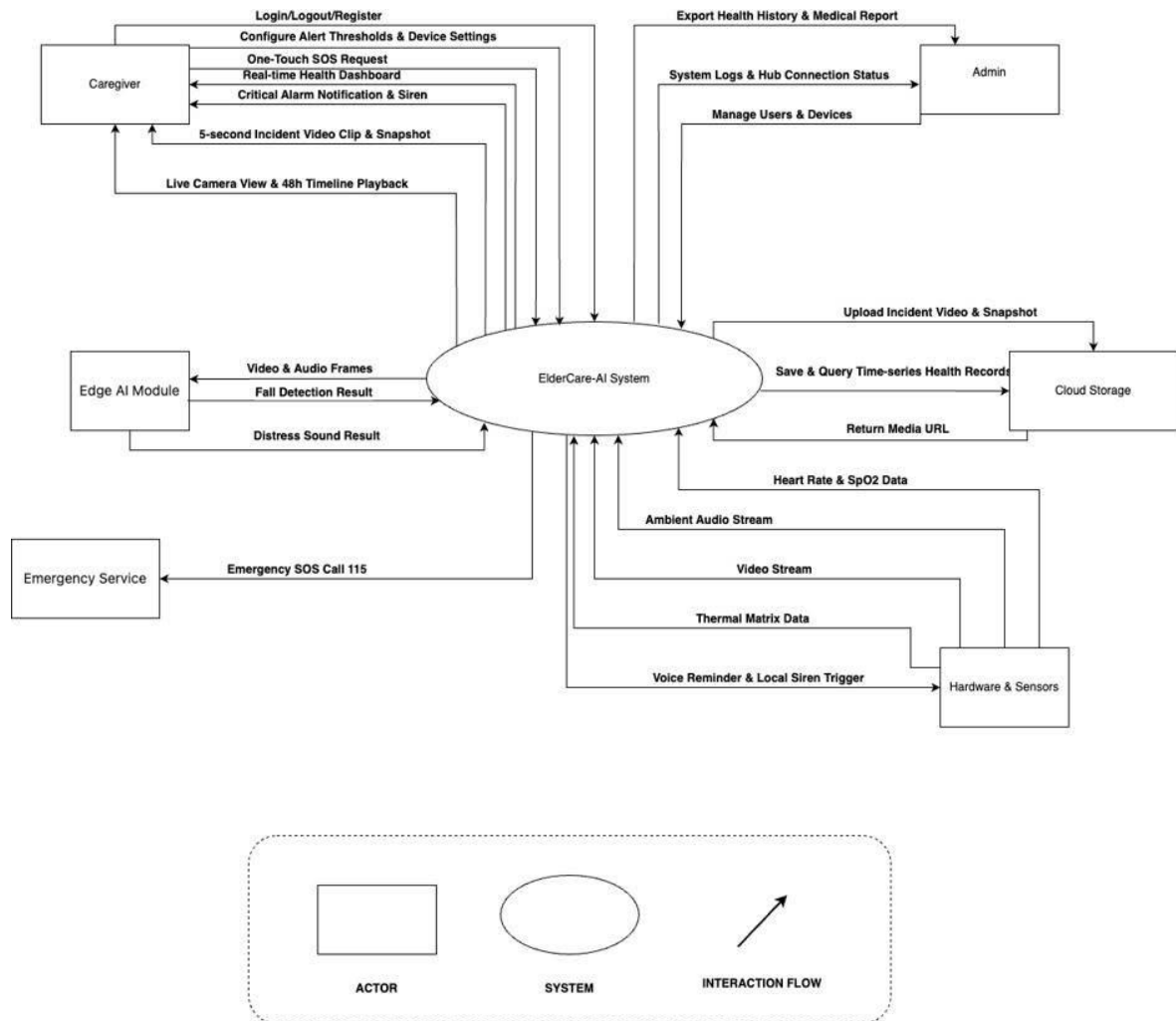
	Camera, AMG8833, BLE Band)			
3.2.2	Convert YOLOv8-Pose ONNX to Deploy Orange Pi 5 NPU	6 days	02/10/2026	07/10/2026
3.2.3	Implement Driver AMG8833 (I2C) & BLE GATT Client (Bleak for Smartband)	6 days	08/10/2026	13/10/2026
3.2.4	Develop Multimodal Sensor Fusion Engine & Rolling Buffer (5s Incident Clip)	5 days	14/10/2026	18/10/2026
3.2.5	Integration Test on Hardware (NPU + Thermal + Audio + BLE Band)	3 days	19/10/2026	21/10/2026
3.2.6	Fix Bugs & Calibrate Sensor Thresholds	2 days	22/10/2026	23/10/2026
3.3	Sprint 3 (Media Server, Cloud Storage & Midterm Report)	24 days	24/10/2026	16/11/2026
3.3.1	Setup Media Server on Hub (WebRTC Live Stream & HLS Continuous Recording)	4 days	24/10/2026	27/10/2026
3.3.2	Prepare & Present Midterm Report	3 days	28/10/2026	30/10/2026

3.3.3	Cloud Storage Integration (MinIO / S3 for 5s Incident Clips & Snapshots)	4 days	31/10/2026	03/11/2026
3.3.4	Implement Critical Push Alert & Siren Override (Firebase FCM + Notifee)	4 days	04/11/2026	07/11/2026
3.3.5	Integrate Smart Voice Audio Reminder on Hub (Local Speaker Alert)	3 days	08/11/2026	10/11/2026
3.3.6	End-to-End Test (Hub to Cloud Backend & Storage)	3 days	11/11/2026	13/11/2026
3.3.7	Fix Bugs & Network Latency Tuning	3 days	14/11/2026	16/11/2026
3.4	Sprint 4 (Mobile/Web Client, Field Testing & Deployment)	18 days	17/11/2026	04/12/2026
3.4.1	Build Mobile App (React Native: Dashboard, Live WebRTC, SOS 115 Dial)	5 days	17/11/2026	21/11/2026
3.4.2	Build Web Dashboard (Next.js: Vitals Analytics & Timeline Playback)	4 days	22/11/2026	25/11/2026
3.4.3	End-to-End System Testing & Execute 50 Emergency Test Cases	4 days	26/11/2026	29/11/2026

3.4.4	System Hardening, Systemd Auto-start Daemons & Dockerize Backend	3 days	30/11/2026	02/12/2026
3.4.5	Fix Final Bugs & Deploy System Prototype	2 days	03/12/2026	04/12/2026
4.	Project's Retrospective Meeting	1 day	05/12/2026	05/12/2026
5.	Final Release & Submission	1 day	05/12/2026	06/12/2026

12. System Architecture Overview

Figure 1. System context diagram



13. Potential Risks and Mitigation Strategies

Categories	Risk Name	Potential Risks	Mitigation Strategies
Technology	Sensor Failure / Noise	AMG8833 thermal sensors, obscured cameras, or disconnected/dead BLE bands cause the system to collect insufficient monitoring data.	• Implement a Heartbeat check mechanism and handle sensor errors instantly. • Set up a fallback logic: If 1 sensor is missing, the remaining channels are still active to issue a conformity alert. • Play a voice prompt at the Edge Hub when it detects that the user forgot to wear the BLE ring.
	Edge AI performance on Orange Pi 5 (Edge Performance)	Simultaneous processing of camera frames, thermal matrix, and audio analysis on the Orange Pi 5 may experience overload, lag, or overheating of the NPU.	• Optimize the AI model through conversion to ONNX/RKNN format to get the most out of the Rockchip NPU hardware (RK3588S). • Reduce the input resolution sufficiently (320x320 / 640x640) to ensure a minimum inference speed of 30 FPS.
System	Network & Power Outage	A sudden loss of Internet connection or power outage interrupts the sending of FCM alerts to the relative's app and the Cloud.	• AI processing and 100% local storage of rolling data (RAM/SD Card) at the Edge Hub. • Automatically re-synchronize crash data to the Cloud as soon as the Internet connection is restored. • Configure the service to automatically start (Systemd Daemons) as soon as power is restored without manual intervention.

Security & Privacy	Video & Data Privacy Leaks	Videos of the elderly's personal activities in the home are leaked or illegally accessed via the Internet.	• Never stream raw continuous video to the Cloud; all AI Vision processes 100% at Edge Hub. • Only extracts and encrypts 5-second clip uploads to the Cloud when there is a red alert issue. • Uses TLS/SSL protocols, JWT encryption, strict access permissions (RBAC), and automatically deletes data after the specified deadline.
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14. Expected Outcomes / Deliverables

- **Software:** Complete and deploy software systems with intelligent, easy-to-use interfaces, optimizing user experience (including Edge Gateway Daemon, Sensor Fusion Engine, Cloud Backend, Caregiver Mobile App, and Web Dashboard).
- **Document:** Provides technical documentation and hardware manuals, including Software Requirements Specification (SRS), User Story documentation, architectural design, APIs, installation, and system configuration instructions.
- **Test report:** A detailed report on the results of software and hardware testing, including unit tests, API integration tests, latency benchmarks, detected errors, and field test verification under real-world emergency scenarios.
- **Project progress report:** Provides progress tracking reports and milestones throughout the development process, including Sprint Reviews, Retrospectives, status reports, and midterm presentation evaluations.

- **Source code updates:** The system's source code is stored on the GitHub repository, with full commits, development branches, and containerized deployment scripts (Docker & Docker Compose).

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